



Welcome to A Catholic Introduction to Artificial Intelligence Class 2! The Early History of AI & Automation

"Technology can often remove an inconvenience, but it cannot remove the wound in the human heart." - Fyodor Dostoevsky

Learning Objectives

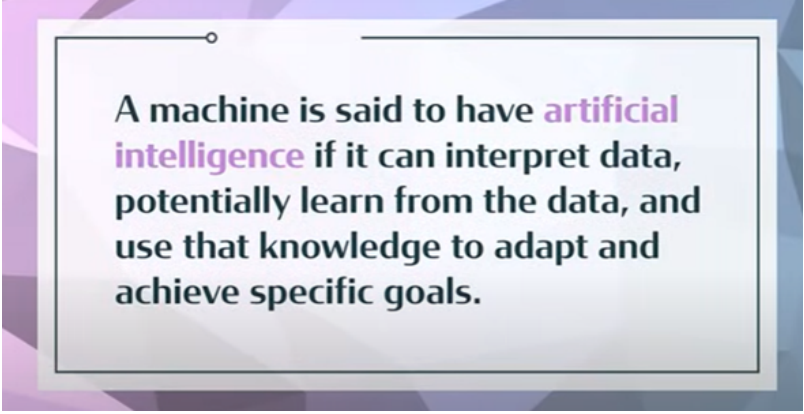
By the end of this class, students will be able to:

1. Describe the historical roots of AI and automation.
2. Identify key figures and milestones in early AI development.
3. A Short History of AI
4. Understand the difference between traditional automation and intelligent systems.
5. Explain the concept of the "AI Winter" and its significance.

1. Class 1 Review

"What is AI?"

Artificial intelligence is an ever-expanding branch of computer science that focuses on making machines capable of performing tasks that usually require human intelligence, such as reasoning, solving problems, making decisions, perceiving the world, making meaning, and learning from experience.

A definition of artificial intelligence presented in a light blue rectangular box with a thin black border. The box is set against a background of purple and blue geometric shapes. The text inside the box is centered and reads: "A machine is said to have **artificial intelligence** if it can interpret data, potentially learn from the data, and use that knowledge to adapt and achieve specific goals." The words "artificial intelligence" are highlighted in a darker blue color.

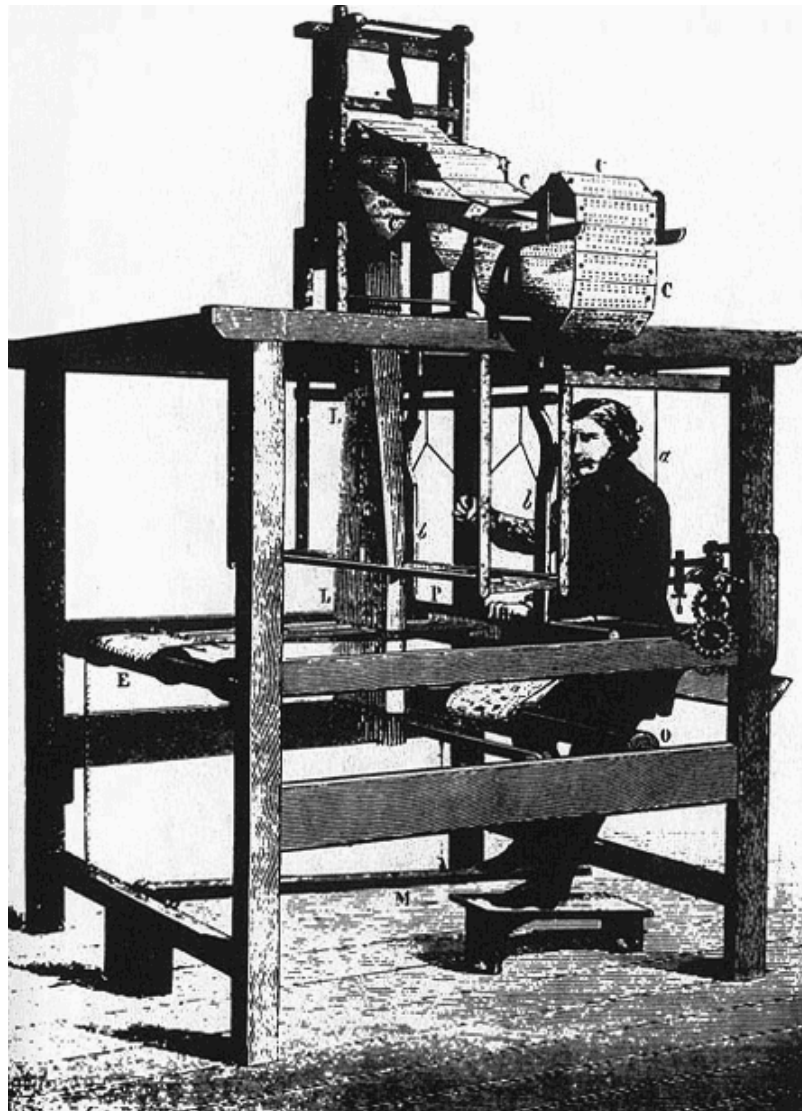
A machine is said to have **artificial intelligence** if it can interpret data, potentially learn from the data, and use that knowledge to adapt and achieve specific goals.

AI is about building **agents** that **perceive** and **act** to achieve goals. Agents perceive its environment through **sensors** and act using **actuators**.

2. Early Automation and the Roots of AI

Introduction

The story of modern artificial intelligence begins long before the invention of the computer. Early machines such as the Jacquard Loom in 1801 demonstrated how mechanical systems could automate repetitive tasks. By using punch cards to control weaving patterns, the loom showed that human creativity could be encoded into a set of instructions a machine could execute. A century later, assembly lines refined this idea on an industrial scale. Henry Ford's moving assembly line didn't "think," but it mechanized repetitive labor so efficiently that it transformed manufacturing and reshaped society's expectations of what machines could do. These early innovations reveal a long-standing human desire to offload routine work onto tools that can perform it faster, more consistently, and without fatigue.



The leap from mechanical automation to computational intelligence emerged in the 20th century, most notably through the work of Alan Turing. Turing proposed that machines could, in principle, perform any logical operation a human could, provided they were given the right instructions. His famous Turing Test reframed the question of machine intelligence: instead of asking whether a machine “thinks,” he asked whether its behavior is indistinguishable from that of a human conversational partner. This idea continues to influence how we evaluate AI systems today

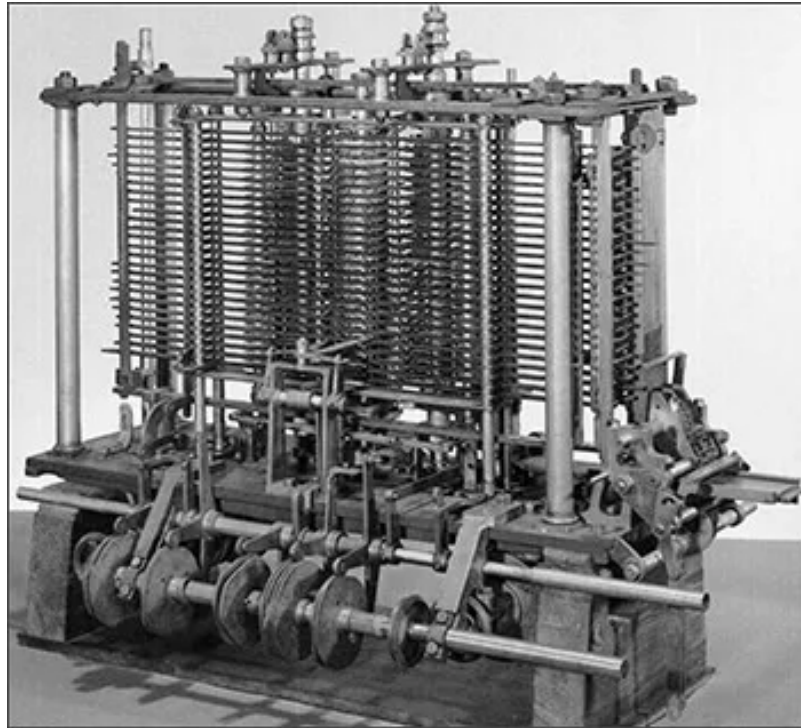
A Clear Timeline from Loom to Turing

1801 — Jacquard Loom

Joseph Marie Jacquard introduces a loom that uses **punch cards** to control weaving patterns. This is the first widely adopted example of a *programmable machine*. It doesn't compute, but it follows encoded instructions — a huge conceptual leap.

1830s — Babbage & Ada Lovelace

Charles Babbage designs the **Analytical Engine**, the first mechanical general-purpose computer concept. Ada Lovelace writes what we now call the **first computer program**, explicitly referencing Jacquard's punch-card idea as inspiration.



1890 — Hollerith Tabulating Machine

Herman Hollerith uses punch cards to automate the U.S. Census. This is the direct ancestor of IBM and early 20th-century data processing.



1912 — Alan Turing is born

More than a century after Jacquard's invention.



1936 — Turing’s “On Computable Numbers”

Turing formalizes the idea of a universal machine that can compute anything describable by rules — the theoretical foundation of modern computing.

1950 — The Turing Test

Turing proposes a behavioral test for machine intelligence: if a machine’s responses are indistinguishable from a human’s, it can be considered intelligent.

Summary

[Who Invented the Computer](#)

3. The First Generation of AI and its Challenges.

Early AI: Key Figures, Milestones, and the Foundations of Symbolic Reasoning

The early development of artificial intelligence emerged from a blend of mathematical theory, cognitive science, and computer engineering. Several key figures shaped the field’s foundations. **Alan Turing** is often considered the intellectual father of AI for proposing, in 1950, that machines could exhibit intelligent behavior. His “Imitation Game,” now known as the **Turing Test**, reframed intelligence as behavior rather than

internal consciousness. Another foundational figure, **John McCarthy**, coined the term *artificial intelligence* in 1956 and organized the Dartmouth Conference, the event widely regarded as the birth of AI as a formal research discipline. Researchers such as **Marvin Minsky**, **Allen Newell**, and **Herbert Simon** further advanced the field by developing early problem-solving programs and articulating the idea that human reasoning could be modeled computationally.

One of the earliest practical demonstrations of machine “intelligence” was **ELIZA**, created by Joseph Weizenbaum in 1966. ELIZA was a text-based conversational program that simulated a psychotherapist by using simple pattern-matching rules. When a user typed a sentence, ELIZA applied scripted transformations—such as turning “I feel...” into “Why do you feel...?”—to create the illusion of understanding. Although ELIZA had no genuine comprehension, it revealed how easily humans attribute meaning and emotion to machine responses. Around the same time, researchers developed **expert systems**, programs designed to replicate the decision-making abilities of human specialists. Systems like **MYCIN**, which diagnosed bacterial infections, used structured rules and knowledge bases to produce recommendations. These systems demonstrated that computers could perform narrow, high-level reasoning tasks previously thought to require human expertise.

Underlying both ELIZA and expert systems was the dominant paradigm of early AI: **symbolic AI**, also known as rule-based or classical AI. Symbolic AI assumes that intelligence arises from manipulating symbols—words, numbers, logical statements—according to explicit rules. In this view, thinking is a form of structured problem-solving: if the system has the right rules and the right representations, it can reach correct conclusions. Symbolic AI systems typically rely on **if-then rules**, logical operators, and hierarchical knowledge structures. This approach excelled in well-defined domains such as mathematics, medical diagnosis, and formal reasoning, where human experts could articulate the rules governing their decisions.

However, symbolic AI also revealed its limitations. Human reasoning is often intuitive, contextual, and difficult to reduce to explicit rules. As problems grew more complex and ambiguous, rule-based systems became brittle, requiring enormous effort to maintain and expand. These challenges eventually opened the door to statistical and learning-based approaches, which now dominate modern AI.

Still, the early era of symbolic AI laid essential groundwork. It provided the first working models of machine reasoning, introduced the idea of computers as partners in intellectual tasks, and established the conceptual vocabulary—rules, representations, inference—that continues to influence AI research today.

<https://www.masswerk.at/elizabot/>

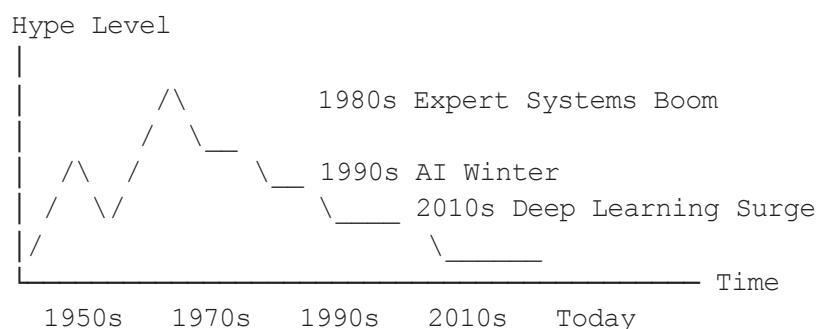
4. Explain the concept of the "AI Winter" and its significance.

AI Winter: Causes, Consequences, and Why Today Is Different

The history of artificial intelligence is marked not only by breakthroughs but also by periods of deep disappointment known as **AI Winters**. An *AI Winter* refers to a time when enthusiasm, funding, and research progress in AI sharply declined. These downturns occurred primarily because early AI systems failed to meet the ambitious expectations set by researchers, the media, and government agencies. In the 1960s and 1970s, many believed that human-level reasoning, language understanding, and robotics were only a decade away. When these predictions proved unrealistic, confidence collapsed. Funding agencies such as DARPA cut support, companies abandoned AI projects, and the field entered a prolonged period of stagnation.

A major cause of these winters was the pattern of **overpromising and underdelivering**. Early AI researchers often made bold claims about what their systems could achieve. Programs like SHRDLU, which manipulated blocks in a simulated world, or early machine translation systems seemed impressive in narrow contexts. But when applied to real-world complexity, they broke down. Expert systems in the 1980s suffered a similar fate: they worked well in controlled environments but were brittle, expensive to maintain, and unable to adapt. As limitations became clear, the gap between hype and reality widened, leading to disillusionment. This cycle repeated more than once, creating multiple AI Winters across the late 20th century.

A simple way to visualize this history is through a **hype-over-time chart**:



This curve shows the recurring pattern: rapid excitement, inflated expectations, collapse, and eventual renewal.

What makes today different is that modern AI rests on foundations that earlier generations lacked. First, we now have **massive datasets** generated by the internet, sensors, smartphones, and global digital infrastructure. Early AI researchers had only tiny, hand-crafted datasets, which limited what their systems could learn. Second, today's hardware is exponentially more powerful. Graphics processing units (GPUs), tensor processing units (TPUs), and specialized accelerators allow modern neural networks to train on billions of examples. This computational scale was unimaginable during previous AI eras. Finally, breakthroughs in deep learning, transformer architectures, and self-supervised learning have enabled systems to generalize far beyond the narrow rule-based approaches of the past.

While hype still exists, today's AI systems deliver practical, measurable results across language, vision, science, and industry. The field is no longer driven by speculation alone but by demonstrable capability. These structural differences make another AI Winter less likely, even as researchers remain cautious about repeating the mistakes of overpromising.

5. Exit Ticket

Artificial intelligence has gone through several major shifts since its earliest days, and understanding these changes helps explain both past challenges and today's progress. Early AI researchers in the 1950s and 1960s believed that computers would soon match human reasoning. Key figures such as Alan Turing, John McCarthy, Marvin Minsky, and Herbert Simon laid the foundations for the field, but their predictions often exceeded what technology could realistically achieve at the time.

These overly optimistic expectations led to periods known as *AI Winters*. An AI Winter occurs when excitement and funding collapse because AI systems fail to deliver on the promises made by researchers or the media. This happened more than once: early language-processing systems struggled with real-world complexity, and later expert systems in the 1980s proved too brittle and expensive to maintain. Each time, the gap between hype and actual performance caused disappointment and slowed progress.

A major theme across these winters was the cycle of *overpromising and underdelivering*. Researchers showcased impressive demos in narrow settings, but those systems could not generalize to broader tasks. As limitations became clear, governments and companies withdrew support. This pattern created a repeating rise-and-fall curve in AI enthusiasm, often described as the AI hype cycle.

Today's AI boom is different because modern systems are powered by enormous datasets, far more powerful hardware, and new learning techniques such as deep learning and transformers. Instead of relying on hand-written rules, today's models learn patterns from billions of examples. This combination of data, computing power, and new algorithms allows AI to deliver real, measurable results across language, vision, science, and industry—making another AI Winter less likely, even as researchers remain cautious about repeating past mistakes.

6. Vocabulary

1. **AI Winter** — A period when funding and enthusiasm for AI research sharply declined due to unmet expectations.
2. **Expert System** — An early AI program that used rules and knowledge bases to mimic human specialists.
3. **ELIZA** — A 1960s chatbot that used pattern-matching to simulate conversation without true understanding.

4. **Symbolic AI** — An approach to AI that uses rules, logic, and symbols to represent knowledge.
5. **Rule-Based System** — A program that makes decisions using explicit “if-then” statements.
6. **Knowledge Base** — A structured collection of facts and rules used by expert systems.
7. **Inference Engine** — The component of an expert system that applies rules to draw conclusions.
8. **Machine Learning** — A modern AI approach where systems learn patterns from data rather than relying on hand-written rules.
9. **Neural Network** — A computational model inspired by the brain, used in today’s deep learning systems.
10. **Training Data** — Large datasets used to teach modern AI models how to recognize patterns.